NEXUS · DOCUMENTATION PACKAGE · DELIVERABLE 11

Deployment Guide

How the API and app are actually deployed today (local), the container path the repository supports, and labelled recommendations for public hosting.

Document control

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Evidence labels used in this document: Implemented verified in code, configuration or test output; Inferred reasoned from evidence but not executed; Recommendation proposed, not implemented; Open question needs a decision; Unknown no evidence available; Estimate planning figure, not a measurement.

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1 Current deployment position

> **No hosting platform is configured.** The repository contains no provider-specific configuration: no cloud manifests, infrastructure-as-code or deploy workflow. The README states that public deployment "still needs a hosted API URL" and that the earlier production hostname "was not a configured service" Implemented. Sections 3 and 4 document what the repository supports, and section 5 contains recommendations only.

Supported deployment targets

ID	Target	Defined by	Verified
DEP-01	Local Node process (development)	`backend/package.json` `start` / `dev`, `docs/ONLINE_OFFLINE.md`	Server module loaded and tested in process (16/16) Implemented; listening server not started for this package
DEP-02	Single container with a named volume	`backend/Dockerfile`, `backend/compose.yaml`	Not executed Unknown; CI smoke test never succeeded (CI-02)
DEP-03	Root `docker-compose.yml`	Repository root	Inferred Will not persist data correctly. Do not use (AUD-02)
DEP-04	Android debug/release builds	`app/build.gradle.kts`, Android CI	Debug build last green in CI on 2026-09-24; release never built in the retrieved history

2 Prerequisites

- API: Node.js 20 (CI) or 22 (Docker image); npm. For containers: Docker with Compose v2.
- App: Android Studio with SDK Platform 37; JDK 17+ (the build uses a 17 toolchain); an emulator or device on API 24+.
- A strong `JWT_SECRET` of 32+ characters, stored outside git (`.env*` files are git-ignored).

3 Run the API locally (DEP-01)

```
# PowerShell, from backend/
npm.cmd ci
$env:JWT_SECRET = node -e "console.log(require('crypto').randomBytes(48).toString('base64url'))"
npm.cmd start          # listens on http://localhost:5263
curl.exe http://localhost:5263/health
```

- Data is written to `backend/nexus-data.json` unless `DATABASE_PATH` is set. The file is git-ignored.
- Keep the same secret across restarts, or every session becomes invalid (RB-06).
- Optional test accounts: run `docs/seed_test_users.ps1` against a **local** API only (AUD-14).

3.1 Point the app at the local API

Debug build targets

Device	Command

Device	Command
Emulator	Run from Android Studio, or `.\gradlew.bat :app:assembleDebug`. The default URL `http://10.0.2.2:5263/api/` reaches the host.
Phone on the same LAN	`.\gradlew.bat :app:assembleDebug -PNEXUS_API_URL=http://<your-LAN-IP>:5263/api/`, and allow port 5263 through the firewall.

The URL must end in `/` and match `https?://host[:port]/path/`, or the Gradle build fails at configuration time (TR-07).

4 Run the API in a container (DEP-02)

```
# from backend/, with JWT_SECRET exported in the shell or a git-ignored .env file
docker compose up --build -d
docker compose ps          # STATUS should become "healthy"
curl http://localhost:5263/health
```

- The image is `node:22-alpine`. It installs production dependencies only, runs as the non-root `node` user, exposes 5263, and stores data at `/data/nexus-data.json` on the `nexus-data` named volume.
- A Docker `HEALTHCHECK` calls `/health` every 30 s. `restart: unless-stopped` is set.
- Compose refuses to start if `JWT_SECRET` is unset (`${JWT_SECRET:?…}`).
- Run **one** replica only (CON-01). Back up the volume (RB-04).

5 Public hosting (recommendations)

Everything in this section is **Recommendation**. It describes the minimum the code requires from any host; it does not select a provider.

Host requirements derived from the code

ID	Requirement	Why
DEP-R1	Runs a Node 20+ process or an OCI container, with exactly one instance	In-memory state and single-file store (CON-01)
DEP-R2	Persistent disk or volume mounted at `/data` (or `DATABASE_PATH`) that survives redeploys	Ephemeral file systems would lose all data
DEP-R3	HTTPS termination with a valid certificate and a stable domain	Release builds reject HTTP (NFR-02)
DEP-R4	Secret store for `JWT_SECRET`; `NODE_ENV=production`	Startup check; production error masking
DEP-R5	Health check against `/health`	Liveness
DEP-R6	Configure Express `trust proxy` before go-live	Rate limiting behind the proxy (AUD-06, IMP-06)
DEP-R7	Fix AUD-01 and set up scheduled backups before real users	Data-loss risk

5.1 Building a release that uses the hosted API

1. Confirm `https://<host>/health` responds.

2. Locally: `.\gradlew.bat :app:assembleRelease -PNEXUS_API_URL=https://<host>/api/` ; or set the GitHub repository variable `NEXUS_API_URL` so CI builds the release APK and AAB (CI-01).
3. Sign the artifacts. Signing is not configured in the repository (AUD-08).
4. Complete the release checklist (REL-CHECK).

6 Rollback

No versioned deployments exist, so rollback is manual **Implemented**. **Recommendation** Tag each deployed commit. To roll back, redeploy the previous tag against the same data volume. Data written by a newer version stays in the file (the JSON store has no schema version), so take a backup before every deploy.